

# Topic: Fourier Series and Differential Equations

## I. Introduction

In this section we will explore one of the ways Fourier series can be used to find solutions of differential equations.

In particular, we will continue to investigate differential equations of the form:

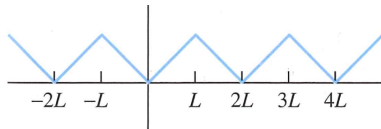
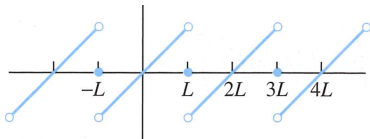
$$y^{(M)} + a_{M-1}y^{(M-1)} + a_{M-2}y^{(M-2)} + \cdots + a_1y' + a_0y = f(t) \quad (1)$$

New Assumption:  $f(t)$  is a **periodic function of  $t$**

Note: Since we are only making assumptions for the function on the right hand side of (1), only the **particular** solution ( $y_p$ ) is changed.

The **homogeneous** solution ( $y_h$ ) on the other hand, is exactly the same.

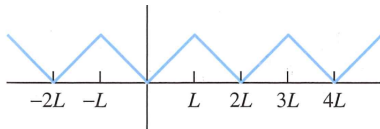
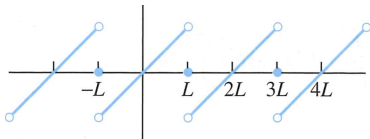
Suppose  $f(t)$  looked like one of the following:



The next question is: What does  $y_p(t)$  look like? That is, what would be our guess for  $y_p(t)$ ?

Based on the work we did in the previous section, then  $y_p(t)$  should be **similar to  $f(t)$** .

Unfortunately, there is a serious problem with this. Why?



Periodic functions are often not differentiable at certain points (see above). So choosing  $y_p(t)$  similar to  $f(t)$  would mean that  $y_p$  would also not be differentiable.

Therefore, it wouldn't make sense to plug it into, say  $y'' - 4y = f(t)$ , since we can't compute  $y_p''$

Luckily, since  $f(t)$  is periodic, then it can be written as a Fourier Series which is differentiable everywhere!

Therefore, the guess for  $y_p(t)$  would also be a Fourier Series.

Let the Fourier Series for  $f(t)$  be given by: 
$$f(t) = \sum_{n=-\infty}^{\infty} f_n e^{i\pi n t/L}$$

Let the Fourier Series for  $y_p(t)$  be given by: 
$$y_p(t) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n t/L}$$

For every differential equation, the Fourier coefficients ( $f_n$  and  $d_n$ ) are related.

It turns out that if we know  $f_n$  we can find  $d_n$ .

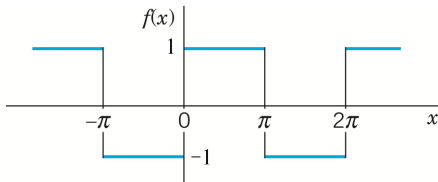
Knowing  $y_n$  is the same as knowing the solution to the differential equation.

Ex: Solve the differential equation:

$$y' + 2y = f(t)$$

where  $f(t)$  is given by the figure.

First, we need to represent  $f$  by a differentiable function. That is,



$$f(t) = \sum_{n=-\infty}^{\infty} f_n e^{i\pi n t/L} \quad (\text{with } L = \pi)$$

$$f(x) = \begin{cases} -1 & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases} \quad f(x+2\pi) = f(x)$$

The Fourier coefficients of  $f(t)$  are  $f_n = \frac{1 - (-1)^n}{i\pi n} \implies f(t) = \sum_{n=-\infty}^{\infty} \frac{1 - (-1)^n}{i\pi n} e^{int}$

(Note :  $\sum = \sum_{n=-\infty}^{\infty}$ )

(1) Guess:  $y_p(t) = \sum d_n e^{int}$

(2) Compute:  $y_p'(t) = \frac{d}{dt} \left[ \sum d_n e^{int} \right] = \sum d_n \frac{d}{dt} [e^{int}] = \sum d_n i n e^{int}$

(3) Plug:  $y_p' + 2y_p = f(t)$

$$\sum d_n i n e^{int} + 2 \sum d_n e^{int} = \sum f_n e^{int}$$

(4) Group:  $\sum (in + 2) d_n e^{int} = \sum f_n e^{int}$

(5) Solve:  $(in + 2) d_n = f_n \implies d_n = \frac{1}{in + 2} f_n = \frac{1}{in + 2} \frac{1 - (-1)^n}{i\pi n} = \frac{1 - (-1)^n}{i\pi n(in + 2)}$

$$\implies y_p(t) = \sum_{n=-\infty}^{\infty} \frac{1 - (-1)^n}{i\pi n(in + 2)} e^{int} \quad \text{and} \quad y_h(t) = c_1 e^{2t} \implies y(t) = c_1 e^{2t} + \sum_{n=-\infty}^{\infty} \frac{1 - (-1)^n}{i\pi n(in + 2)} e^{int}$$



Ex: Consider the following second order differential equation:

$$a_2 y'' + a_1 y' + a_0 y = f(t)$$

where  $f(t) = f(t + 2L)$ .

Derive the relationship between Fourier coefficients of  $f(t)$  and  $y_p(t)$

Let

$$f(t) = \sum_{n=-\infty}^{\infty} f_n e^{i\pi n t/L} \quad \text{and}$$

$$y_p(t) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n t/L} \quad \Rightarrow \quad y_p'(t) = \sum_{n=-\infty}^{\infty} \left(\frac{i n \pi}{L}\right) d_n e^{i\pi n t/L} \quad \Rightarrow \quad y_p''(t) = \sum_{n=-\infty}^{\infty} \left(\frac{i n \pi}{L}\right)^2 d_n e^{i\pi n t/L}$$

Plugging in  $a_2 y_p'' + a_1 y_p' + a_0 y_p = f(t)$

$$a_2 \sum_{n=-\infty}^{\infty} \left(\frac{i n \pi}{L}\right)^2 d_n e^{i\pi n t/L} + a_1 \sum_{n=-\infty}^{\infty} \left(\frac{i n \pi}{L}\right) d_n e^{i\pi n t/L} + a_0 \sum_{n=-\infty}^{\infty} d_n e^{i\pi n t/L} = \sum_{n=-\infty}^{\infty} f_n e^{i\pi n t/L}$$

$$\sum_{n=-\infty}^{\infty} \left[ a_2 \left(\frac{i n \pi}{L}\right)^2 + a_1 \left(\frac{i n \pi}{L}\right) + a_0 \right] d_n e^{i\pi n t/L} = \sum_{n=-\infty}^{\infty} f_n e^{i\pi n t/L}$$

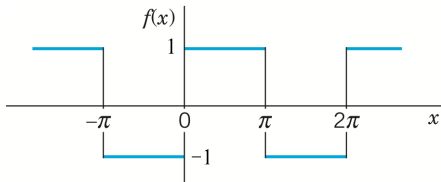
$$\left[ a_2 \left(\frac{i n \pi}{L}\right)^2 + a_1 \left(\frac{i n \pi}{L}\right) + a_0 \right] d_n = f_n \quad \Rightarrow \quad d_n = \left[ a_2 \left(\frac{i n \pi}{L}\right)^2 + a_1 \left(\frac{i n \pi}{L}\right) + a_0 \right]^{-1} f_n$$

$$d_n = \left[ \frac{L^2}{-a_2 \pi^2 n^2 + a_1 \pi L i n + a_0 L^2} \right] f_n$$

Ex: Verify this result with the previous example:

$$y' + 2y = f(t)$$

where  $f(t)$  is given by the figure.



In the previous example we found  $f_n = \frac{1 - (-1)^n}{i\pi n}$  and  $d_n = \frac{1 - (-1)^n}{i\pi n(in + 2)}$

and for this eqn:  $a_2 = 0$  ,  $a_1 = 1$  ,  $a_0 = 2$  ,  $L = \pi$

$$d_n = \left[ \frac{L^2}{-a_2\pi^2 n^2 + a_1\pi L i n + a_0 L^2} \right] f_n = \left[ \frac{\pi^2}{-(0)\pi^2 n^2 + (1)\pi^2 i n + 2\pi^2} \right] f_n = \left[ \frac{\pi^2}{\pi^2 i n + 2\pi^2} \right] f_n = \left[ \frac{1}{i n + 2} \right] f_n$$

$$d_n = \left[ \frac{1}{i n + 2} \right] \frac{1 - (-1)^n}{i\pi n} = \frac{1 - (-1)^n}{i\pi n(in + 2)} \quad \checkmark$$

Ex: Solve the equation:  $y'' + 2y' + y = \sin t$  for  $t \in [-\pi, \pi]$

$$\left( \text{Note: } \sin t = \frac{e^{it} - e^{-it}}{2i} \right)$$

$$\sin t = \frac{1}{2i}e^{it} + \frac{-1}{2i}e^{-it} = \frac{1}{2i}e^{i(1)t} + \frac{-1}{2i}e^{i(-1)t} \leftarrow \text{Fourier Series where } f_n = \begin{cases} \frac{1}{2i} & n = 1 \\ \frac{-1}{2i} & n = -1 \\ 0 & n \neq \pm 1 \end{cases}$$

Assuming  $y_p(t) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n t/L}$  and by the previous result with:  $a_2 = 1$ ,  $a_1 = 2$ ,  $a_0 = 1$ ,  $L = \pi$

$$d_n = \left[ \frac{L^2}{-a_2 \pi^2 n^2 + a_1 \pi L n + a_0 L^2} \right] f_n = \left[ \frac{\pi^2}{-\pi^2 n^2 + 2\pi^2 i n + \pi^2} \right] f_n = \left[ \frac{1}{-n^2 + 2i n + 1} \right] f_n$$

$$d_n = \begin{cases} \left[ \frac{1}{2i} \right] f_1 & n = 1 \\ \left[ \frac{-1}{2i} \right] f_{-1} & n = -1 \\ 0 & n \neq \pm 1 \end{cases} \implies d_n = \begin{cases} -\frac{1}{4} & n = 1 \\ -\frac{1}{4} & n = -1 \\ 0 & n \neq \pm 1 \end{cases}$$

$$\implies y_p(t) = \sum_{n=-\infty}^{\infty} d_n e^{int} = d_1 e^{i\pi(1)t/L} + d_{-1} e^{i(-1)t} = -\frac{1}{4}e^{it} - \frac{1}{4}e^{-it} = -\frac{1}{2}\cos(t)$$

$$\text{Char Eqn: } r^2 + 2r + 1 = (r+1)(r+1) \implies y_h(t) = c_1 e^{-t} + c_2 t e^{-t}$$

$$y(t) = c_1 e^{-t} + c_2 t e^{-t} - \frac{1}{4}e^{it} - \frac{1}{4}e^{-it} = c_1 e^{-t} + c_2 t e^{-t} - \frac{1}{2}\cos(t)$$

## II. Even and Odd Extensions

Defn: Suppose  $f(t)$  is defined on  $(0, L)$ .

(1) The even extension of  $f(t)$  is defined by

$$f^E(t) = \begin{cases} f(t) & 0 < t < L \\ f(-t) & -L < t < 0 \end{cases}$$

(2) The odd extension of  $f(t)$  is defined by

$$f^O(t) = \begin{cases} f(t) & 0 < t < L \\ -f(-t) & -L < t < 0 \end{cases}$$

Note: This is a handy trick if you have a function that is on defined for  $0 < t < L$  and want to find its Fourier series. Moreover, since we are creating an even or odd function, we can utilize the Even/Odd properties to:

(1) Simplify our calculations and

(2) Choose to write our series in terms of either **cosine** or **sine**.



### III. Purely Cosine and Sine Fourier Series

When working with Fourier Series, the following results are often useful:

- (1) If  $f(x)$  is an even function, its Fourier Series can be written purely in terms of **cosine**.

$$f(t) = \sum_{n=-\infty}^{\infty} f_n e^{i\pi n t/L} = \frac{a_0}{2} + \sum_{n=1}^{\infty} 2f_n \cos\left(\frac{\pi n t}{L}\right) \quad (\text{cosine series})$$

$$a_n = 2f_n = 2 \left[ \frac{1}{L} \int_0^L f(x) \cos\left(\frac{\pi n x}{L}\right) dx \right] = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{\pi n x}{L}\right) dx$$

- (2) If  $f(x)$  is an odd function, its Fourier Series can be written purely in terms of **sine**.

$$f(t) = \sum_{n=-\infty}^{\infty} f_n e^{i\pi n t/L} = \sum_{n=1}^{\infty} 2if_n \sin\left(\frac{\pi n t}{L}\right) \quad (\text{sine series})$$

$$b_n = 2if_n = 2i \left[ \frac{-i}{L} \int_0^L f(x) \sin\left(\frac{\pi n x}{L}\right) dx \right] = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{\pi n x}{L}\right) dx$$

Ex: Let  $f(t) = t$  be defined for  $0 < t < 2$ . (1) Determine the even and odd extension of  $f(t)$   
(2) Find the corresponding cosine and sine series

(1) Note:  $L = 2$  and  $f(-t) = -t$

$$f^E(t) = \begin{cases} t & 0 < t < 2 \\ -t & -2 < t < 0 \end{cases}$$

$$f^E(t) = f^E(t+2)$$

$$f^O(t) = \begin{cases} t & 0 < t < 2 \\ t & -2 < t < 0 \end{cases}$$

$$f^O(t) = f^O(t+2)$$

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$$(2) \quad f_0^E = \frac{1}{2} \int_0^2 t \, dt = 2, \quad f_n^E = \frac{1}{2} \int_0^2 t \cos\left(\frac{n\pi t}{2}\right) dt = \dots = \frac{2}{n^2\pi^2} [(-1)^n - 1]$$

$$\Rightarrow f^E(t) = f_0^E + \sum_{n=1}^{\infty} 2f_n^E \cos\left(\frac{n\pi t}{2}\right) = 2 + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2} [(-1)^n - 1] \cos\left(\frac{n\pi t}{2}\right)$$

$$f_n^O = \frac{-i}{2} \int_0^2 t \sin\left(\frac{n\pi t}{2}\right) dt = \dots = \frac{-2i}{n\pi} (-1)^{n+1}$$

$$\Rightarrow f^O(t) = \sum_{n=1}^{\infty} 2if_n^O \sin\left(\frac{n\pi t}{2}\right) = \sum_{n=1}^{\infty} \frac{4}{n\pi} (-1)^{n+1} \sin\left(\frac{n\pi t}{2}\right)$$